Measuring Media Slant through Image Analysis

Christian Arnold
University of Birmingham

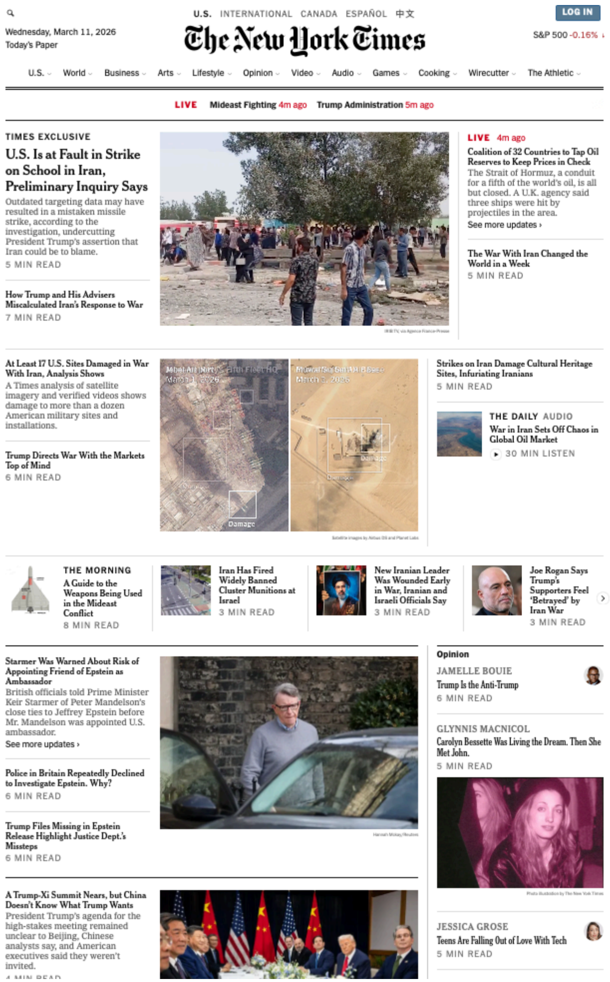
Andreas Küpfer
Technical University of Darmstadt

Oliver Rittmann
University of Mannheim, MZES

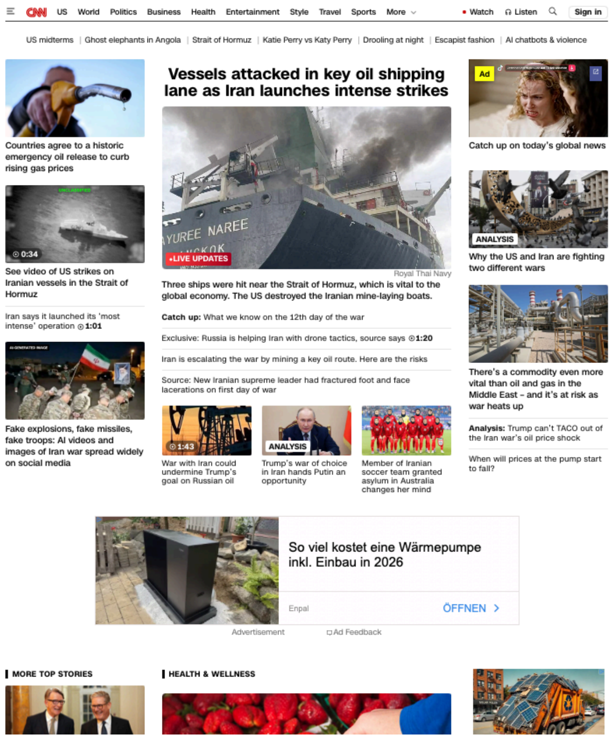
Michelle Torres
University of California, Los Angeles

Annual Meeting of the European Political Science Society · June 18–20, 2026 · Belfast, UK

Images Dominate News Websites



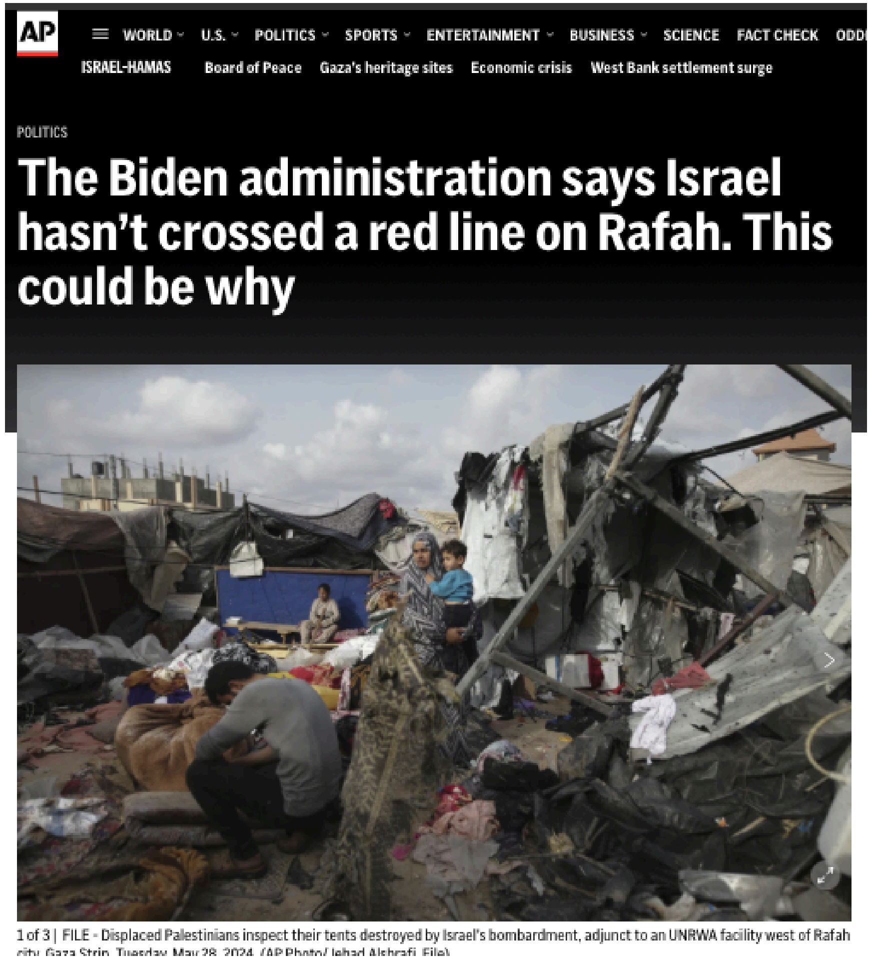
New York Times, March 11, 2026



CNN, March 11, 2026

Bild, March 11, 2026

The Same Event, Pictured Differently



AP News, June 1, 2024

Guardian, May 29, 2024

Images Are Central in Modern Media Environments, Yet Unmeasured

Images Are Central in Modern Media Environments, Yet Unmeasured

- Images are omnipresent in news media, particularly online

Images Are Central in Modern Media Environments, Yet Unmeasured

- Images are omnipresent in news media, particularly online
- Images shape the perceptions of events, potentially more forcefully than text

Images Are Central in Modern Media Environments, Yet Unmeasured

- Images are omnipresent in news media, particularly online
- Images shape the perceptions of events, potentially more forcefully than text
- We know about textual media slant based, but what about visual reporting?

Images Are Central in Modern Media Environments, Yet Unmeasured

- Images are omnipresent in news media, particularly online
- Images shape the perceptions of events, potentially more forcefully than text
- We know about textual media slant based, but what about visual reporting?
- Difficult to answer due to methodological hurdles

This Study

Our goal:
*Develop a method to **measure media slant** based on the **visual reporting** of media outlets on specific events.*

This Study

Our goal:
*Develop a method to **measure media slant** based on the **visual reporting** of media outlets on specific events.*

1. We develope a two-step, **unsupervised framework** for scaling images

This Study

Our goal:
*Develop a method to **measure media slant** based on the **visual reporting** of media outlets on specific events.*

1. We develope a two-step, **unsupervised framework** for scaling images

2. Application: Visual reporting on the Gaza war, 10 outlets, Oct 2023–Oct 2024

This Study

Our goal:
*Develop a method to **measure media slant** based on the **visual reporting** of media outlets on specific events.*

1. We develope a two-step, **unsupervised framework** for scaling images
2. **Application:** Visual reporting on the Gaza war, 10 outlets, Oct 2023–Oct 2024
3. **Evidence** for media slant in visual reporting, particularly for right-wing media

Framework

Framework: Two Steps

Core assumption: *News outlets that are close to each other in the latent space will, on average, use more similar news imagery.*

Framework: Two Steps

> ***Core assumption:*** *News outlets that are close to each other in the latent space will, on average, use more similar news imagery.*

1. **Measurement of image similarity**: Encode each image with a pre-trained Vision Transformer (ViT)
   → extract the `[CLS]` token embedding (512 dims)
   → compute pairwise **cosine similarities** across all image dyads

Framework: Two Steps

> ***Core assumption:*** *News outlets that are close to each other in the latent space will, on average, use more similar news imagery.*

1. **Measurement of image similarity**: Encode each image with a pre-trained Vision Transformer (ViT)
   → extract the `[CLS]` token embedding (512 dims)
   → compute pairwise **cosine similarities** across all image dyads
2. **Statistical model to analyse image similarity**: Additive-Multiplicative Effects (AME) network model to recover latent positions of
   (a) **media outlets** based on the news imagery used by them and
   (b) **images** themselves

Step 2: The AME Model

Additive-multiplicative effects *(Hoff 2005, 2021)* on outlet level:

$$\operatorname{atanh}\Big(\operatorname{sim}(\vec{E}_i, \vec{E}_j)\Big) = \mu + \alpha_{O_i} + \alpha_{O_j} + \vec{S}_{O_i}^{\top}\vec{S}_{O_j} + \epsilon_{ij}$$

Step 2: The AME Model

Additive-multiplicative effects *(Hoff 2005, 2021)* on outlet level:

$$\operatorname{atanh}\left(\operatorname{sim}(\vec{E}_i, \vec{E}_j)\right) = \mu + \alpha_{O_i} + \alpha_{O_j} + \vec{S}_{O_i}^{\top}\vec{S}_{O_j} + \epsilon_{ij}$$

$\operatorname{sim}(\vec{E}_i, \vec{E}_j)$ is the **cosine similarity** between the ViT embeddings of images i and j.

Step 2: The AME Model

Additive-multiplicative effects *(Hoff 2005, 2021)* on outlet level:

$$\operatorname{atanh}\Big(\operatorname{sim}(\vec{E}_i, \vec{E}_j)\Big) = \mu + \alpha_{O_i} + \alpha_{O_j} + \vec{S}_{O_i}^{\top}\vec{S}_{O_j} + \epsilon_{ij}$$

$\vec{S}_{O_i}$: Latent position of outlet O that posted image i

$\vec{S}_{O_j}$: Latent position of outlet O that posted image j

Step 2: The AME Model

Additive-multiplicative effects *(Hoff 2005, 2021)* on outlet level:

$$\operatorname{atanh}\left(\operatorname{sim}(\vec{E}_i, \vec{E}_j)\right) = \mu + \alpha_{O_i} + \alpha_{O_j} + \vec{S}_{O_i}^{\top}\vec{S}_{O_j} + \epsilon_{ij}$$

$\vec{S}_{O_i}$: Latent position of outlet O that posted image i

$\vec{S}_{O_j}$: Latent position of outlet O that posted image j

Configuration	$S_i \cdot S_j$	Predicted image similarity
Same pole $(+,+)$ or $(-,-)$	> 0	above baseline
Opposite poles $(+,-)$	< 0	below baseline

Image-Level Model

Assign a position directly to each images rather than outlets:

$$\operatorname{atanh}\left(\operatorname{sim}(\vec{E}_i, \vec{E}_j)\right) = \mu + \vec{S}_i^\top \vec{S}_j + \epsilon_{ij}, \quad \epsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$$

$\vec{S}_i$ **Latent scale position of image** i

Image-Level Model

Assign a position directly to each images rather than outlets:

$$\operatorname{atanh}\left(\operatorname{sim}(\vec{E}_i, \vec{E}_j)\right) = \mu + \vec{S}_i^\top \vec{S}_j + \epsilon_{ij}, \quad \epsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$$

$\vec{S}_i$ **Latent scale position of image i**

- Estimated in a Bayesian framework

Image-Level Model

Assign a position directly to each images rather than outlets:

$$\operatorname{atanh}\Big(\operatorname{sim}(\vec{E}_i, \vec{E}_j)\Big) = \mu + \vec{S}_i^\top \vec{S}_j + \epsilon_{ij}, \quad \epsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$$

$\vec{S}_i$ **Latent scale position of image** i

- Estimated in a Bayesian framework
- No labelled data required

Case Study: Gaza War, 2023–2024

Data: Visual Reporting on the Gaza War

Corpus

Data: Visual Reporting on the Gaza War

Corpus

- Around 17,000 photographs

Data: Visual Reporting on the Gaza War

Corpus

- Around 17,000 photographs
- 10 news outlets: mostly US (e.g, CNN, Fox News) with some international (e.g., Al Jazeera English, Irish Times)

Data: Visual Reporting on the Gaza War

Corpus

- Around 17,000 photographs
- 10 news outlets: mostly US (e.g, CNN, Fox News) with some international (e.g., Al Jazeera English, Irish Times)
- Source: NewsAPI + top article image

Data: Visual Reporting on the Gaza War

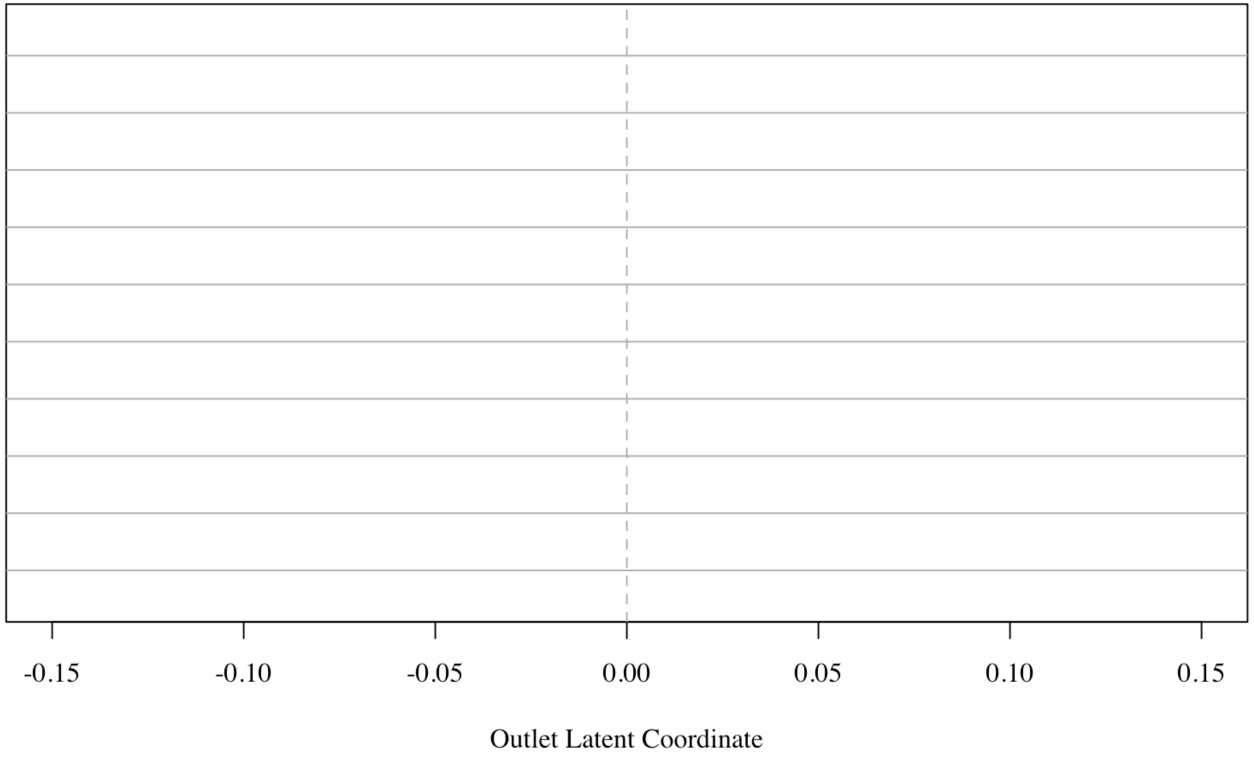
Corpus

- Around 17,000 photographs
- 10 news outlets: mostly US (e.g, CNN, Fox News) with some international (e.g., Al Jazeera English, Irish Times)
- Source: NewsAPI + top article image

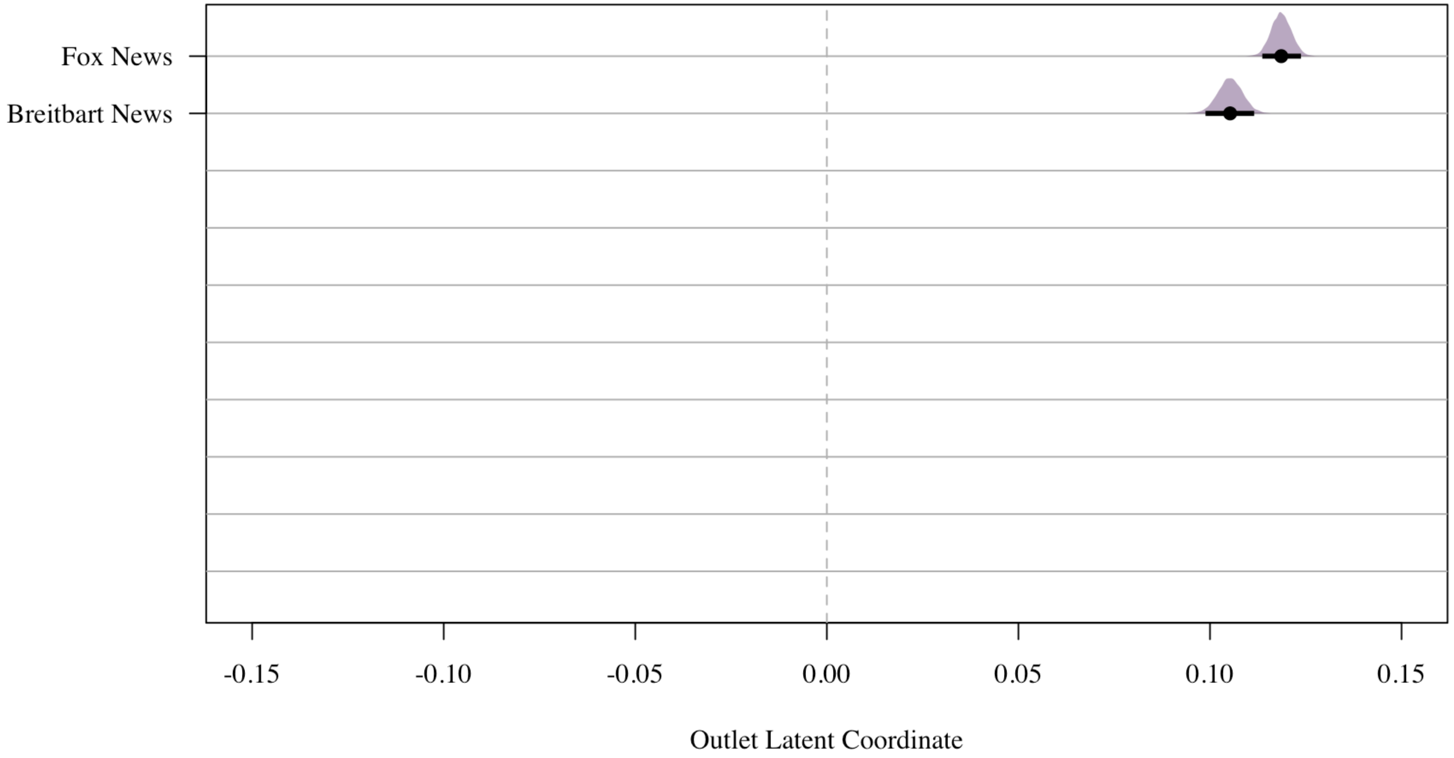
> ***Leading Question:*** *Can we recover media slant exclusively based on news imagery?*

Results

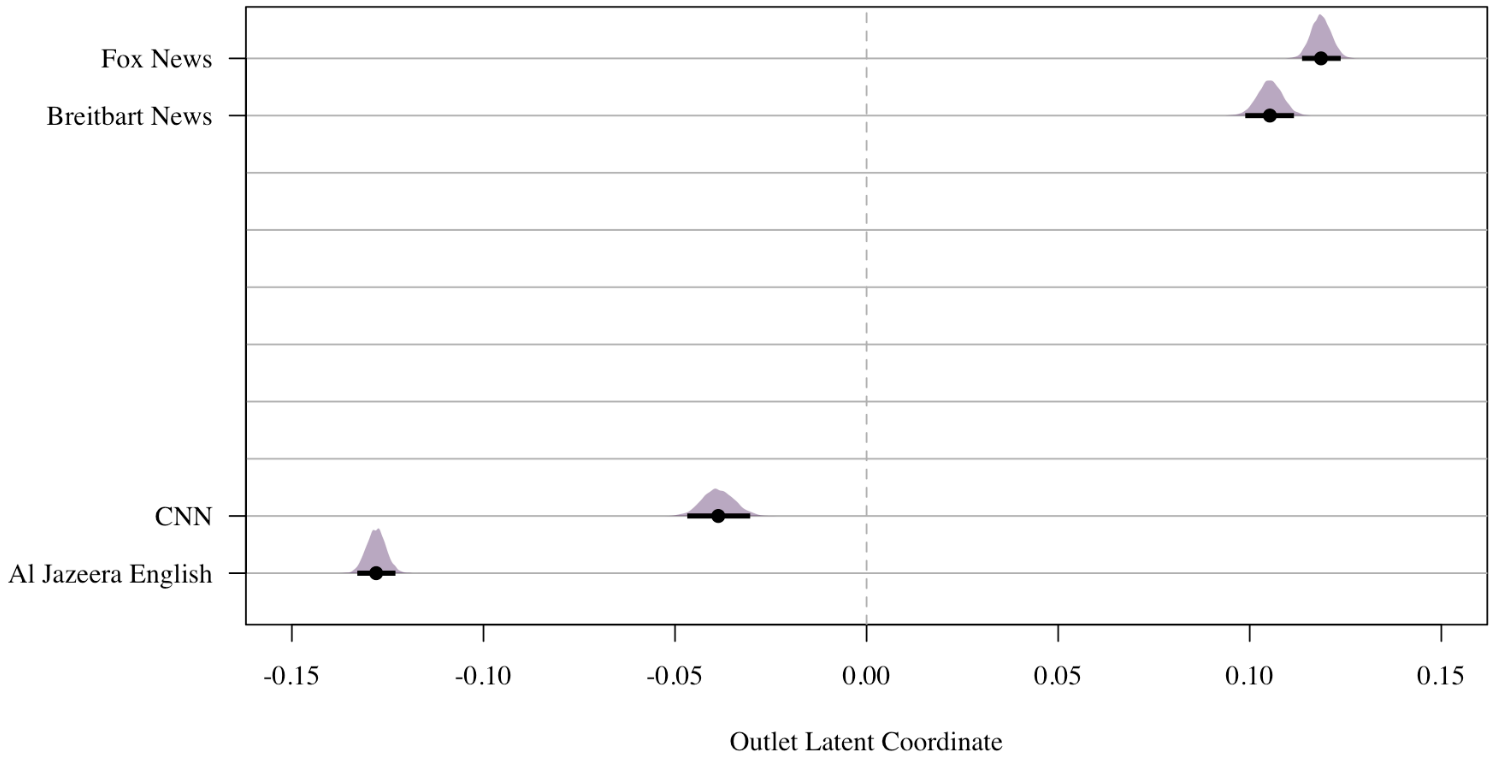
Outlet-Level Visual Slant



Outlet-Level Visual Slant



Outlet-Level Visual Slant



Outlet-Level Visual Slant

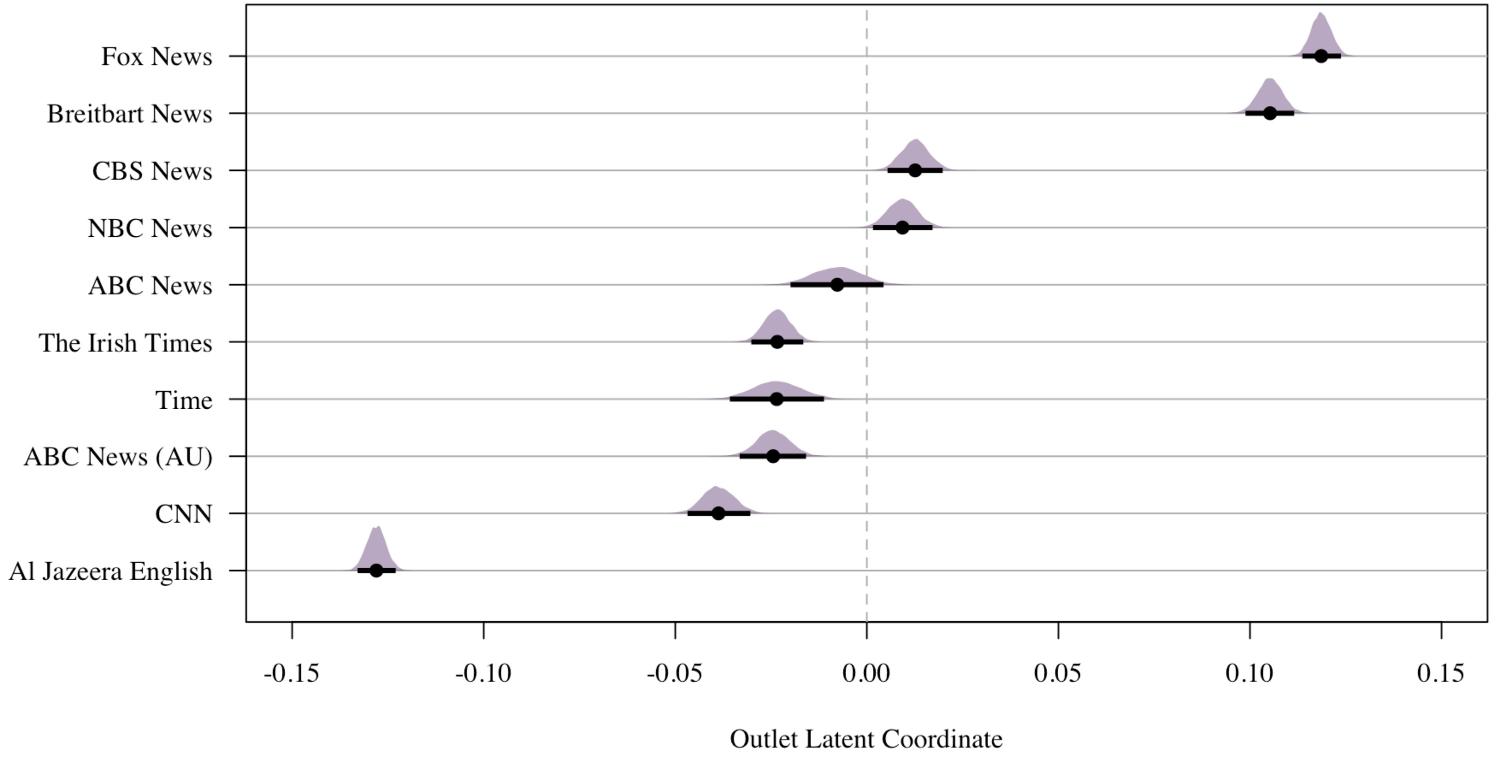


Image-Level: What Is on the Scale?

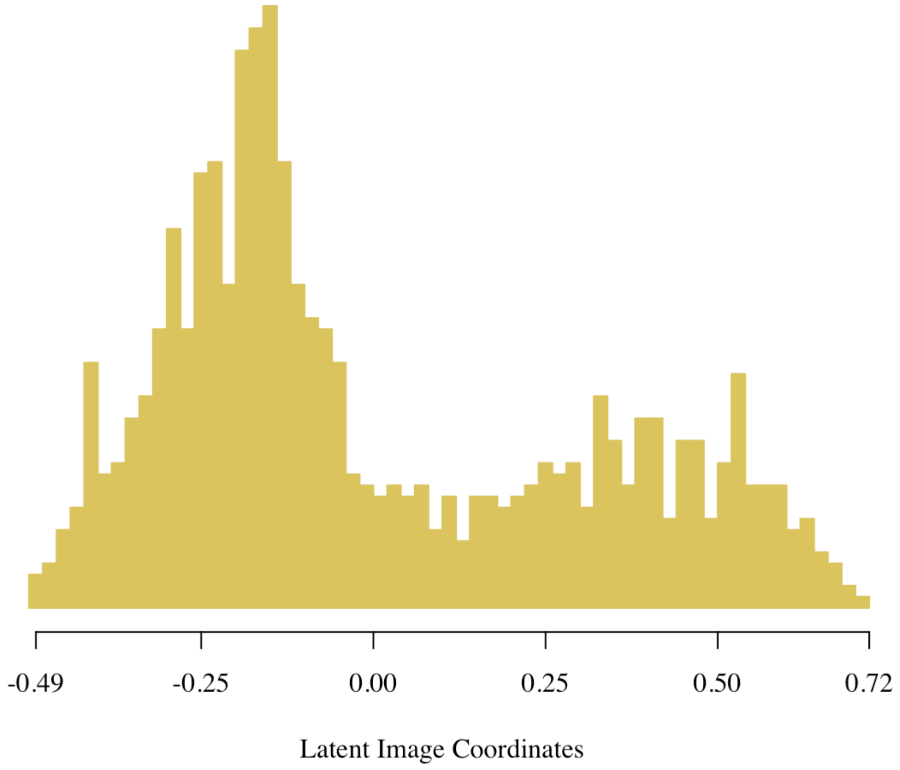


Image-Level: What Is on the Scale?

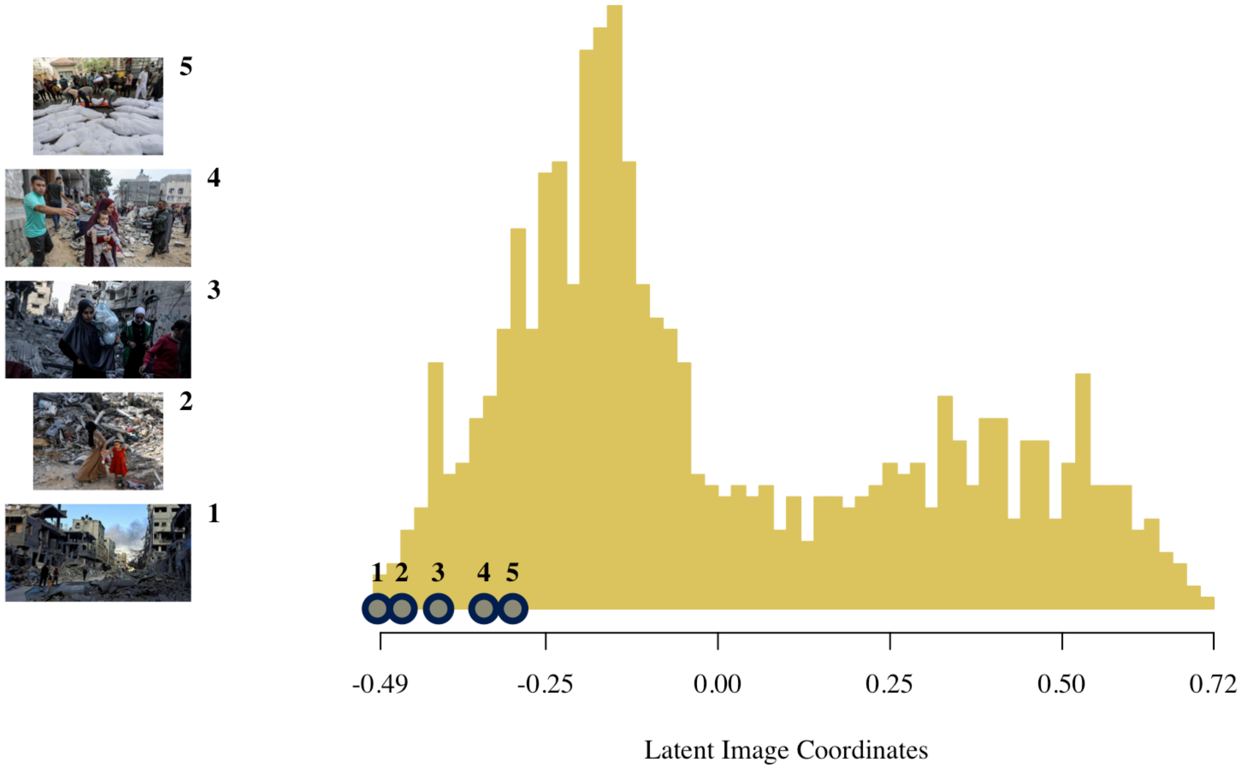
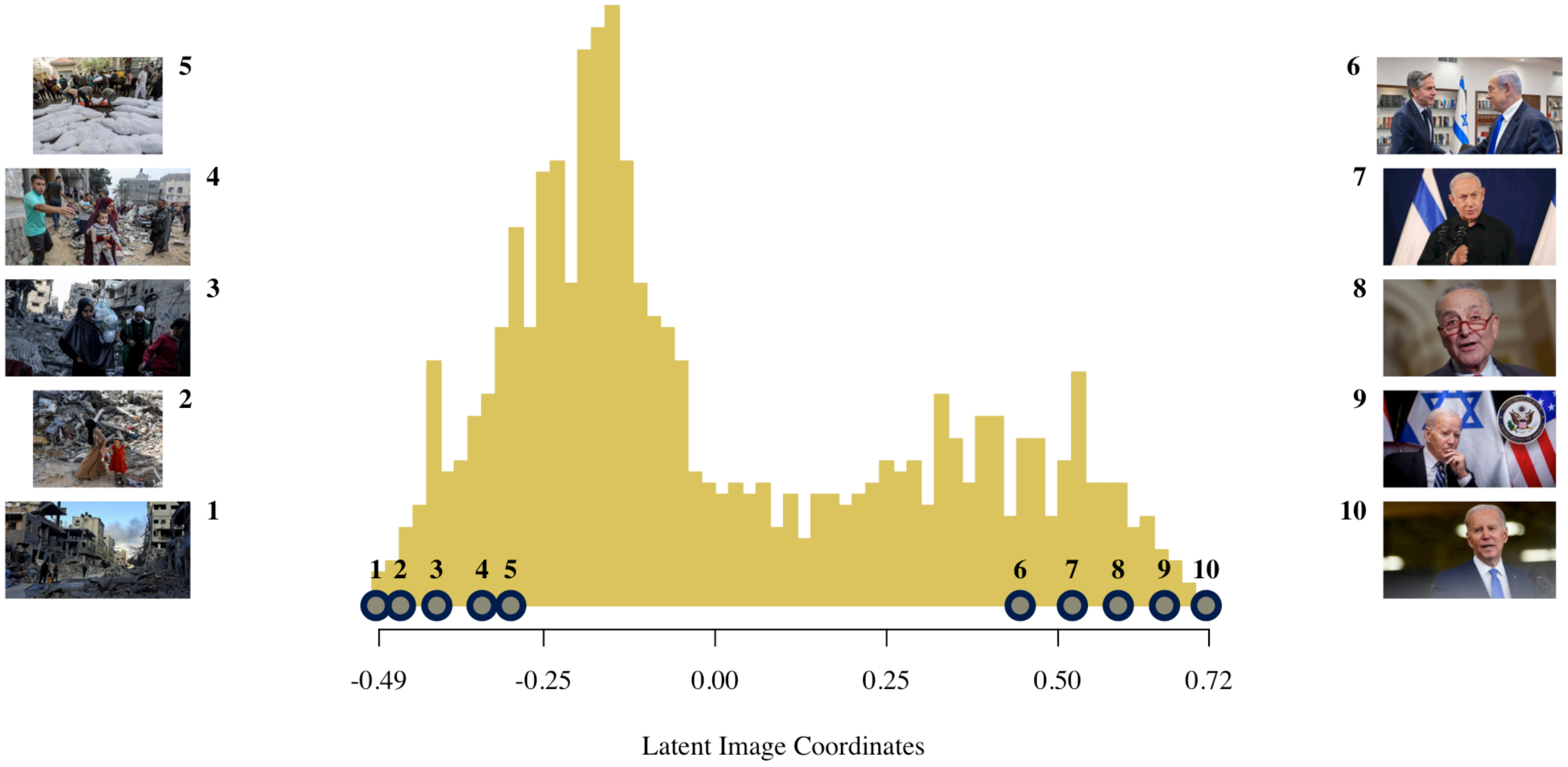


Image-Level: What Is on the Scale?



Is Visual Slant Just Downstream of Text Slant?

> ***Concern:*** *Do outlets differ in which stories they cover (text) or in how they picture the same story?*

Is Visual Slant Just Downstream of Text Slant?

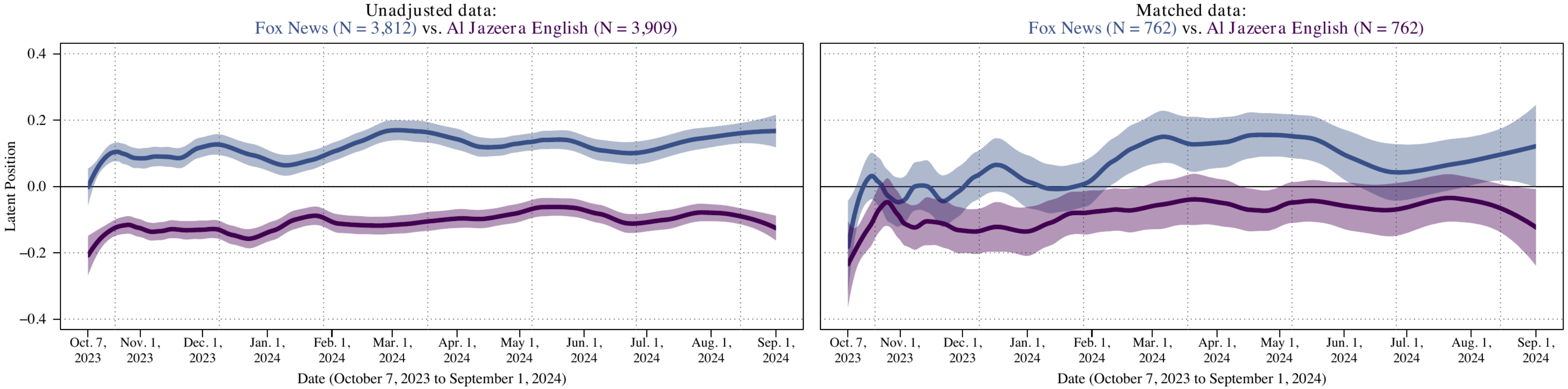
> ***Concern:*** *Do outlets differ in which stories they cover (text) or in how they picture the same story?*

Design: within each day, match articles across two outlets on text-embedding similarity (adapting *Roberts et al. 2020*) and compare images attached to textually similar reports

Is Visual Slant Just Downstream of Text Slant?

> ***Concern:*** *Do outlets differ in which stories they cover (text) or in how they picture the same story?*

Design: within each day, match articles across two outlets on text-embedding similarity (adapting *Roberts et al. 2020*) and compare images attached to textually similar reports



Separation narrows but persist across outlet pairs: visual slant reflects deliberate image selection, not merely topic coverage

Conclusion

Conclusion and Future Directions

Contribution

Conclusion and Future Directions

Contribution

- Scalable framework for recovering latent positions from visual corpora

Conclusion and Future Directions

Contribution

- Scalable framework for recovering latent positions from visual corpora
- Method to measure visual slant in news media

Conclusion and Future Directions

Contribution

- Scalable framework for recovering latent positions from visual corpora
- Method to measure visual slant in news media

Findings on Gaza War coverage

Conclusion and Future Directions

Contribution

- Scalable framework for recovering latent positions from visual corpora
- Method to measure visual slant in news media

Findings on Gaza War coverage

- Far-right outlets share a distinctive visual repertoire; Al Jazeera diverges sharply

Conclusion and Future Directions

Contribution

- Scalable framework for recovering latent positions from visual corpora
- Method to measure visual slant in news media

Findings on Gaza War coverage

- Far-right outlets share a distinctive visual repertoire; Al Jazeera diverges sharply
- Scale captures a substantively meaningful dimension: elite framing vs. humanitarian suffering

Conclusion and Future Directions

Contribution

- Scalable framework for recovering latent positions from visual corpora
- Method to measure visual slant in news media

Findings on Gaza War coverage

- Far-right outlets share a distinctive visual repertoire; Al Jazeera diverges sharply
- Scale captures a substantively meaningful dimension: elite framing vs. humanitarian suffering
- Visual slant persists after matching on text

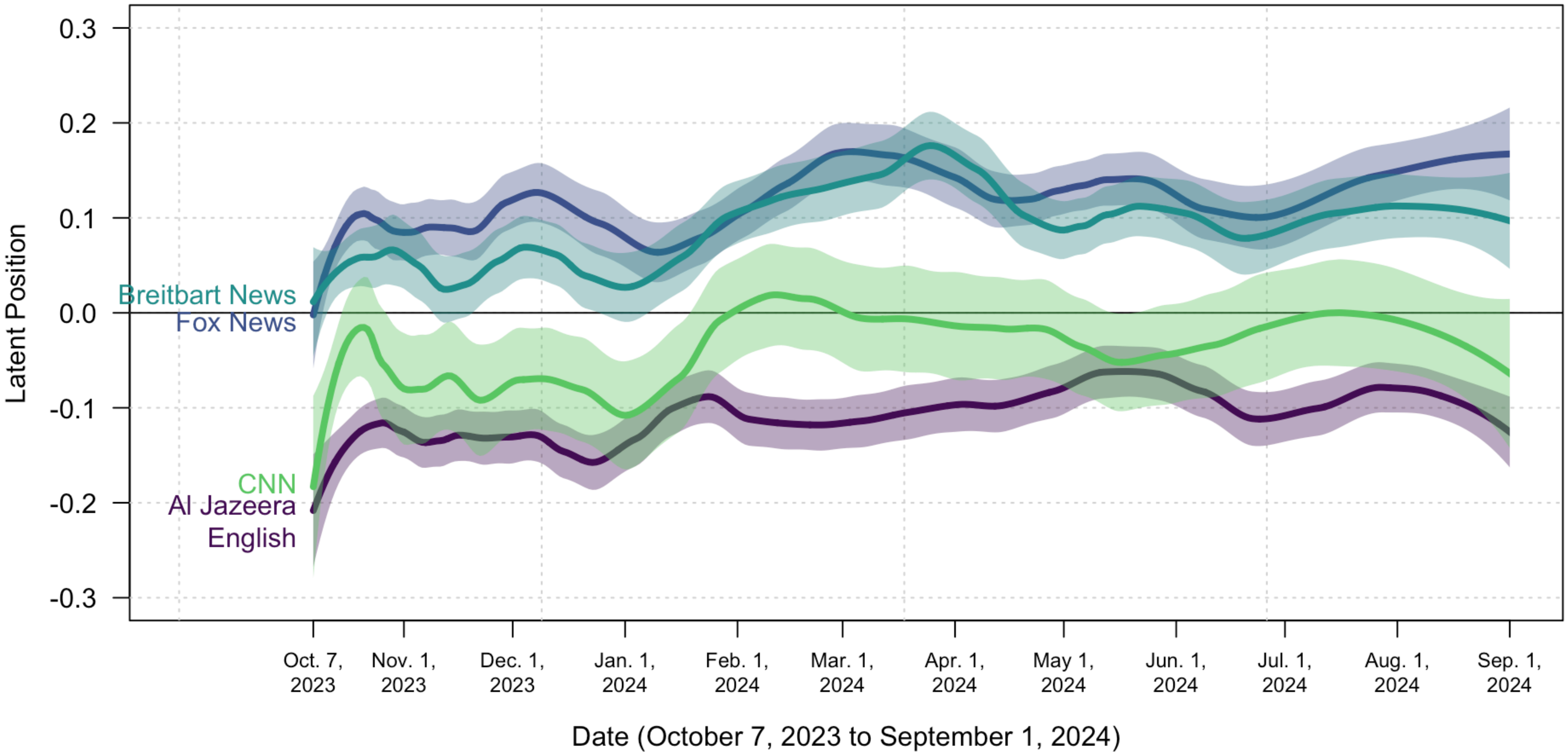
Measuring Media Slant through Image Analysis

Christian Arnold · Andreas Kuepfer · Oliver Rittmann · Michelle Torres

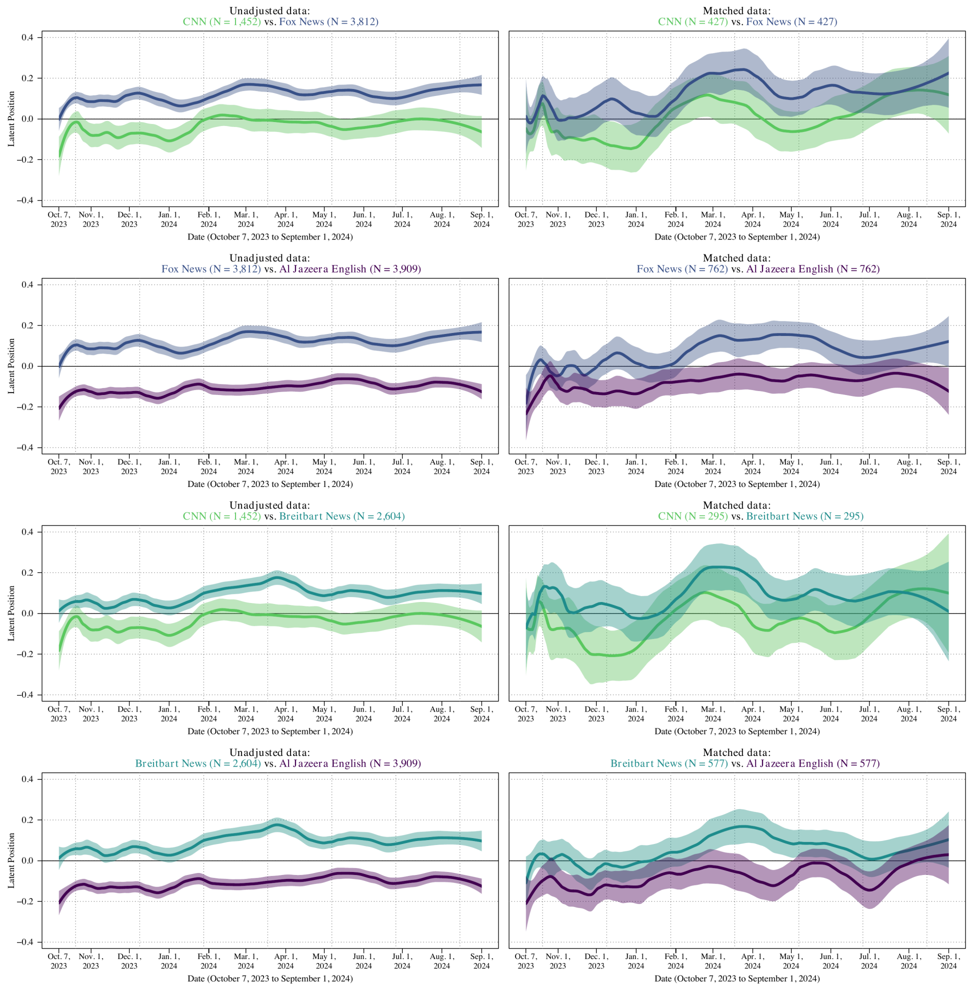
Contact:
andreaskuepfer.github.io
andreas.kuepfer@tu-darmstadt.de
ankuepfer@bsky.social

Appendix

Appendix: Dynamic Visual Slant: Stable Sorting, Parallel Dynamics



Appendix: Visual Slant Conditional on Text — All Pairs



Left: all images per outlet. *Right:* within-day, text-matched article pairs. Separation attenuates but persists.

Appendix: Text-Matching Design

Adapting Roberts et al. (2020) to isolate visual framing from textual content:

Appendix: Text-Matching Design

Adapting Roberts et al. (2020) to isolate visual framing from textual content:

1. Restrict to articles published by both outlets **on the same day**

Appendix: Text-Matching Design

Adapting Roberts et al. (2020) to isolate visual framing from textual content:

1. Restrict to articles published by both outlets **on the same day**

2. Compute pairwise cosine distances on **768-dim text embeddings**

Appendix: Text-Matching Design

Adapting Roberts et al. (2020) to isolate visual framing from textual content:

1. Restrict to articles published by both outlets **on the same day**

2. Compute pairwise cosine distances on **768-dim text embeddings**

3. Greedy 1:1 nearest-neighbour matching across the two outlets

Appendix: Text-Matching Design

Adapting Roberts et al. (2020) to isolate visual framing from textual content:

1. Restrict to articles published by both outlets **on the same day**

2. Compute pairwise cosine distances on **768-dim text embeddings**

3. Greedy 1:1 nearest-neighbour matching across the two outlets

4. Impose a **caliper** (max permissible distance) — drop insufficiently similar pairs

Appendix: Text-Matching Design

Adapting Roberts et al. (2020) to isolate visual framing from textual content:

1. Restrict to articles published by both outlets **on the same day**

2. Compute pairwise cosine distances on **768-dim text embeddings**

3. Greedy 1:1 nearest-neighbour matching across the two outlets

4. Impose a **caliper** (max permissible distance) — drop insufficiently similar pairs

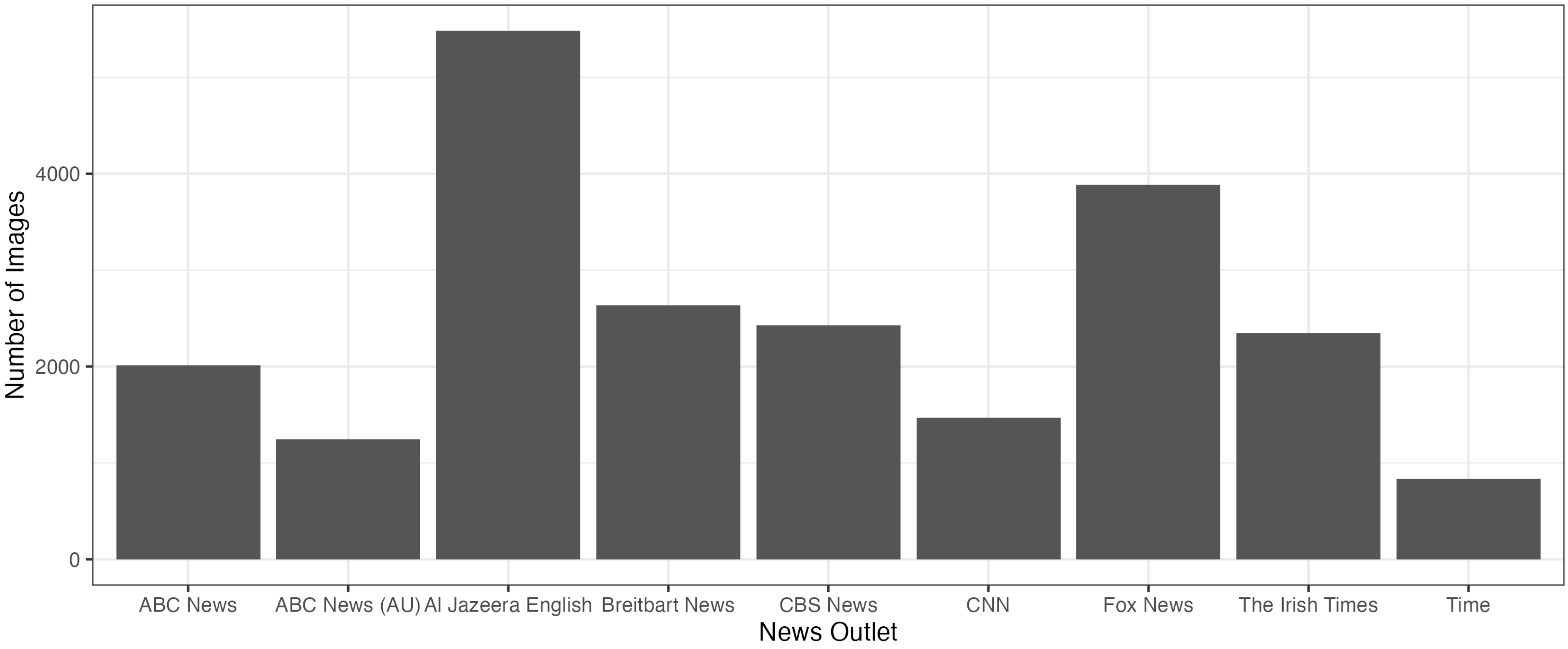
> *Remaining variation in image position between outlets is then attributable to* ***editorial image selection****, not differences in topic coverage.*

Appendix: Text-Matched Differences (Table 1)

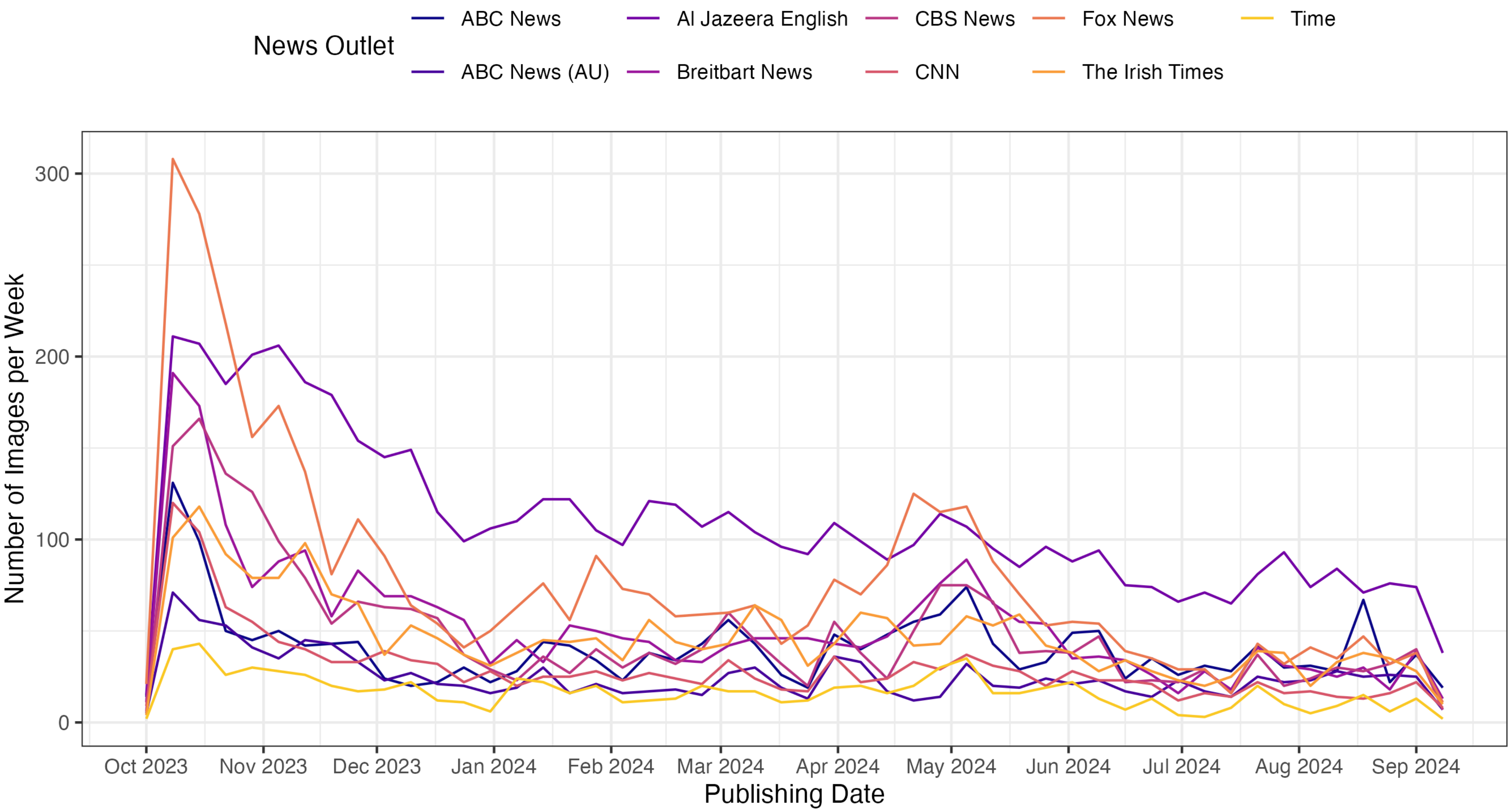
Contrast	Caliper	Est. (SE)	N_{pairs}
CNN vs. Fox News	Unadjusted	0.159 (0.009)	—
	5th pct.	0.102 (0.021)	427
	2.5th pct.	0.092 (0.026)	274
	1st pct.	0.106 (0.032)	176
Fox News vs. Al Jazeera	Unadjusted	0.224 (0.006)	—
	5th pct.	0.137 (0.015)	762
	1st pct.	0.092 (0.023)	287
CNN vs. Breitbart	Unadjusted	−0.128 (0.009)	—
	5th pct.	−0.112 (0.024)	295
	1st pct.	−0.122 (0.040)	73
Breitbart vs. Al Jazeera	Unadjusted	0.191 (0.007)	—
	5th pct.	0.128 (0.015)	577
	1st pct.	0.102 (0.025)	179

OLS of latent image position on outlet, day fixed effects, within-day matched sample.

Appendix: Images per Outlet



Appendix: Images per Outlet over Time



Appendix: Step 2: The AME Model

Additive-multiplicative effects *(Hoff 2005, 2021)* — outlet level:

$$\operatorname{atanh}\Big(\operatorname{sim}(\vec{E}_i, \vec{E}_j)\Big) = \mu + \alpha_{O_i} + \alpha_{O_j} + \vec{S}_{O_i}^{\top}\vec{S}_{O_j} + \epsilon_{ij}$$

μ is the **global baseline** similarity of news imagery across outlet pairs.

Appendix: Step 2: The AME Model

Additive-multiplicative effects *(Hoff 2005, 2021)* — outlet level:

$$\operatorname{atanh}\left(\operatorname{sim}(\vec{E}_i, \vec{E}_j)\right) = \mu + \alpha_{O_i} + \alpha_{O_j} + \vec{S}_{O_i}^{\top} \vec{S}_{O_j} + \epsilon_{ij}$$

α_{O_i}: Random effects for outlet O that posted image i.

α_{O_j}: Random effects for outlet O that posted image j.